Stage 1 Hard Drill

Chapter 2.2 — Concentration Inequalities & Tail Bounds

10 competition-grade questions, full-chapter coverage. Target: under 90 seconds each once the shortcut is spotted.

Q1 — Markov: Choosing the Right Moment

A non-negative loss variable X has $\mathbb{E}[X] = 4$ and $\mathbb{E}[X^3] = 100$ (no information on $\mathbb{E}[X^2]$). Find the Markov bound on $\mathbb{P}(X \geq 10)$ using X directly, and again using X^3. Which is tighter, and by how much?
Hint direction: Markov applies to any non-negative, monotonic transform of X — compare $\mathbb{E}[X]/a$ against $\mathbb{E}[X^3]/a^3$.

Q2 — Markov: Discrete vs. Continuous Achievability

For non-negative X with $\mathbb{E}[X] = \mu$, Markov gives $\mathbb{P}(X \geq 2\mu) \leq \frac{1}{2}$. Construct a distribution that hits this bound **exactly**. Then explain why an absolutely continuous X (one with a genuine density, no point masses) can only **approach** $\frac{1}{2}$ in the limit, never equal it.
Hint direction: equality in Markov requires all the "action" concentrated at a single point — what does a density assign to a single point?

Q3 — Chebyshev: Inverting the Bound

A market maker's inventory has unknown distribution, mean 0, variance 100. Find the smallest symmetric threshold c such that $\mathbb{P}(|\text{Inventory}| \geq c) \leq 0.01$ is guaranteed for **any** distribution with this mean and variance.
Hint direction: solve $1/k^2 = 0.01$ for k, then convert to the original units.

Q4 — Chebyshev: Don't Decompose the Sum

X, Y independent, each mean 0, variance 1. Let $Z = X + Y$. (a) Bound $\mathbb{P}(|Z| \geq 3)$ by applying Chebyshev to Z directly. (b) A classmate instead bounds $\mathbb{P}(|X| \geq 1.5)$ and $\mathbb{P}(|Y| \geq 1.5)$ separately and adds them via a union bound. Compute both numbers and explain which approach is correct and why the other is wasteful.
Hint direction: $\text{Var}(Z) = \text{Var}(X) + \text{Var}(Y)$ by independence — use that fact directly rather than splitting the event.

Q5 — Jensen: How Far Can You Trust the Gap Approximation?

Positive X has $\mathbb{E}[X] = 100$, $\mathbb{E}[X^2] = 10400$ (so $\sigma^2 = 400$). (a) Use Jensen to upper-bound $\mathbb{E}[\ln X]$. (b) Use the second-order Taylor approximation to estimate $\mathbb{E}[\ln X]$ numerically. (c) Can you determine from the given information alone whether the **true** $\mathbb{E}[\ln X]$ is above or below your part-(b) estimate?
Hint direction: part (c) depends on the next Taylor term, which involves a moment you were not given — don't claim more precision than your information supports.

Q6 — Jensen: Volatility Drag

Daily log-return $R \sim \mathcal{N}(\mu, \sigma^2)$. Wealth multiplier is e^R. (a) Use Jensen to show $\mathbb{E}[e^R] > e^{\mathbb{E}[R]}$. (b) Derive the exact value of $\mathbb{E}[e^R]$ via the Normal MGF. (c) Explain how a strategy can have positive arithmetic mean return $\mu > 0$ while its long-run realized (geometric) growth rate is $\mu - \sigma^2/2$, possibly negative.
Hint direction: part (c) is the mechanism behind "volatility drag" — the quantity that compounds over time is not the arithmetic mean.

Q7 — WLLN vs. CLT: the Sample-Size Gap

Per-trade variance $\sigma^2 = 1$ ($\$^2$). Using **Chebyshev/WLLN**, find the minimum n guaranteeing the sample average is within \$0.01 of the true mean with probability $\geq 99\%$. Then find the (much smaller) n that **CLT** says actually suffices for the same target, and compute the ratio of the two.
Hint direction: Chebyshev: $n \geq \sigma^2/(\delta\varepsilon^2)$. CLT: $n \geq (z_{0.005}\,\sigma/\varepsilon)^2$ with $z_{0.005} \approx 2.576$.

Q8 — WLLN: Spot Both Errors

A junior quant argues: "By WLLN, since $n = 20$ independent strategies is large enough, today's average return across all 20 will be very close to the true mean." Identify **two** distinct errors in this reasoning.
Hint direction: one error concerns what "large enough" would even have to be measured against; the other concerns what kind of statement WLLN makes about any single realization, no matter how large n is.

Q9 — CLT: Trusting the Approximation

Daily log-returns have sample mean 0.0003, sample variance 0.0004 ($\sigma = 0.02$), from 10,000 observed days. Using CLT, estimate $\mathbb{P}(\text{avg. log-return over the next 25 days} > 0.01)$. Then discuss whether $n = 25$ is large enough to trust this estimate, given that daily stock returns are known to carry negative skew and excess kurtosis.
Hint direction: standardize using $\sigma/\sqrt{n}$ for the 25-day average; for the trust question, think about which tail (left or right) negative skew actually fattens.

Q10 — Chebyshev vs. CLT: Bound vs. Estimate

1000 i.i.d. fair die rolls: $\mathbb{E}[S] = 3500$, $\text{Var}(S) \approx 2916.67$ ($\sigma \approx 54.0$). (a) Bound $\mathbb{P}(|S - 3500| \geq 200)$ via Chebyshev. (b) Approximate the same probability via CLT. (c) These will differ by roughly two orders of magnitude — explain which one is a rigorous, distribution-free **guarantee**, which is an **asymptotic estimate**, and why a competition grader asking for a "proof of a bound" would reject the CLT number even though it's far more accurate here.
Hint direction: Chebyshev makes no assumption about the shape of the sum's distribution; CLT's accuracy here is a separate fact requiring its own justification (Berry–Esseen).

No solutions above by design — work each closed-book under a 90-second timer, then report which ones took longer. That gap is your next drill target.